

TITUS HYUNKYU LEE

tituslee.com | hvm4sg@virginia.edu | (703) 909-7009 | linkedin.com/in/titushyunkyu | Burke, VA

EDUCATION

University of Virginia, School of Engineering and Applied Science, Charlottesville, VA

August 2023 - May 2027

B.S. in Electrical Engineering and Computer Engineering; Minor in Business

- Cumulative GPA: 3.89/4.0 | A. James Clark Scholar (2023 Cohort)
- Activities: Tau Beta Pi Engineering Honor Society (Professional Development Chair), Theta Tau, Taekwondo Club.
- Study Abroad: UVA in Guatemala (Summer 2024) - Engineering, Public Health, and Development.

RESEARCH & TECHNICAL EXPERIENCE

Software Engineering Intern, **General Atomics**, Herndon, VA

June 2026 - August 2026

- Developed a VS Code extension for inspecting and interpreting raw binary command and telemetry packets, implementing byte-level parsing, packet editing, CRC validation, and database-driven decoding of payload fields and values.
- Automated previously manual embedded flight-software validation by developing a GitLab CI/CD pipeline for code-quality checks, CMake builds, CTest unit testing, JUnit reporting, and build-artifact management.
- Integrated and validated end-to-end command and telemetry paths across embedded development boards and ground-system software using serial, UDP, GNU Radio, and LoRa; documented setup and development workflows for future engineers.

Embedded Systems Engineer (Industry Project), **Interrel**, Dubai, United Arab Emirates

December 2025 - January 2026

- Developed embedded firmware and sensor interfaces on an ESP32 Feather microcontroller with a Bosch BME688 gas sensor, enabling real-time gas classification using a neural-network model trained in Bosch BME AI-Studio.
- Validated real-time clean air vs. smoking/vaping classification on hardware and evaluated system-level integration constraints for Interrel's EOS2 smart thermostat.

Research Intern, **Daegu Gyeongbuk Institute of Science and Technology**, Daegu, South Korea

June 2025 - August 2025

- Designed and fabricated triboelectric nanogenerator (TENG) prototypes (vertical contact-separation and single-electrode), analyzing material performance through surface treatments validated by contact angle measurements.
- Developed a self-powered intruder alert system integrating a TENG sensor with a microcontroller for real-time motion detection; processed signals in LabVIEW and MATLAB and validated performance using an electrometer and oscilloscope.

Teaching Assistant, ECE 2700: Signals & Systems, **University of Virginia**, Charlottesville, VA

August 2025 - Present

- Selected as an undergraduate TA; assisted students with problem sets and core concepts such as Fourier transforms, convolution, and LTI systems through weekly office hours, review sessions, and grading support.

HARDWARE & EMBEDDED PROJECTS

RISC-V CPU Design – Verilog / Quartus / ModelSim

Spring 2026

- Designed and implemented a RISC-V RV32I CPU in Verilog with modular datapath and control logic, progressing to a five-stage pipelined architecture with Fetch, Decode, Execute, Memory, and Writeback stages.
- Implemented register forwarding, load-use stalls, and branch flushes to handle pipeline hazards; verified arithmetic, logic, shifting, memory, jump, and branch functionality using ModelSim waveform analysis and RARS comparison.

Low-Power SRAM Design – Cadence Virtuoso / Spectre

Spring 2026

- Designed and verified a 1 Mb low-power SRAM in FreePDK45 using a hierarchical 16-bank architecture with 6T bitcells, row/column decoders, sense amplifiers, write drivers, pass-gate output muxing, and output latching.
- Reduced active and idle power through selective block enabling and MTCMOS decoder leakage reduction; validated read/write functionality, SNM, timing, and PVT robustness using Cadence Spectre simulations and Ocean scripting.

Embedded Music Player – C / MSPM0

Fall 2025

- Developed an embedded music player on the LP-MSPM0G3507 microcontroller and booster pack; read and applied device datasheets to configure GPIO, timers, and interrupts at the peripheral level for button-controlled playback and pause functionality.
- Implemented a finite state machine (FSM) to manage system states and timing behavior, using hardware timers to sequence pre-coded audio output and validate reliable real-time operation.

Audio Analyzer PCB – KiCad / MultiSim

Spring 2025

- Designed and fabricated a real-time audio analyzer PCB indicating bass and treble frequencies; implemented a signal chain with a summing amplifier, Sallen-Key filters (650 Hz & 2 kHz), and peak detectors.
- Integrated analog and PWM LED drivers using discrete components and MOSFETs to control brightness; verified dynamic LED response through Multisim circuit simulations and live testing with audio input.

SKILLS

- Programming Languages: C, C++, Python, Java, VHDL, Verilog, JavaScript, x86 Assembly.
- Tools & Software: Code Composer Studio, KiCad, Quartus, NI Multisim, Cadence, LabVIEW.
- Relevant Coursework: Signals & Systems, Embedded Systems, VLSI, Computer Architecture & Design, Computer Systems & Organization I & II, Digital Logic Design, Microelectronics, Applied Circuits, Electronics, AI Hardware (In Progress).
- Languages: English (Native), Korean (Native).